Sequences & Series

Section 1: General Binomial Expansion $(1+x)^n$ for Rational n

1 Find the first three terms in the expansion of $(1+4x)^{1/2}$ in ascending powers of x.

a. $1+2x-2x^2$

b. $1+2x+2x^2$

c. $1+4x-8x^2$

d. $1+2x-4x^2$

2 State the range of values of x for which the binomial expansion of $(1+4x)^{1/2}$ is valid.

a. $|x|<\frac{1}{4}$

b. $|x|<4$

c. $|x|<1$

d. $x>0$

3 Expand $(1-2x)^{-1}$ up to the term in x^3.

a. $1+2x+4x^2+8x^3$

b. $1-2x+4x^2-8x^3$

c. $1+2x+2x^2+2x^3$

d. $1-2x-4x^2-8x^3$

Section 2: Approximating Roots via Binomial Series

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4 Use the first three terms of $(1-x)^{1/2} \approx 1 - \frac{1}{2}x - \frac{1}{8}x^2$ with $x = 0.02$ to approximate $\sqrt{0.98}$.

a. 0.98995

b. 0.98990

c. 0.99000

d. 0.98500

5 To approximate $\sqrt{50}$ using the binomial expansion of $(1+x)^{1/2}$, which factorization is most appropriate?

a. $\sqrt{49(1+1/49)} = 7(1+1/49)^{1/2}$

b. $\sqrt{100(1/2)} = 10(1/2)^{1/2}$

c. $(50)^{1/2} = (1+49)^{1/2}$

d. $\sqrt{36+14} = 6(1+14/36)^{1/2}$

6 Find the coefficient of x^2 in the expansion of $\dfrac{1+2x}{(1-x)^2}$.

a. 5

b. 3

c. 4

d. 6

Section 3: Method of Differences & Telescoping Series

7 Express $\dfrac{1}{r(r+1)}$ in partial fractions.

a. $\dfrac{1}{r} - \dfrac{1}{r+1}$

b. $\dfrac{1}{r} + \dfrac{1}{r+1}$

c. $\dfrac{1}{r+1} - \dfrac{1}{r}$

d. $\dfrac{2}{r} - \dfrac{1}{r+1}$

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8 Use the method of differences to find $\sum_{r=1}^{n} \frac{1}{r(r+1)}$.

a. $\frac{n}{n+1}$

b. $1 - \frac{1}{n}$

c. $\frac{n+1}{n}$

d. $1 - \frac{1}{(n+1)^2}$

9 Find the sum to infinity $\sum_{r=1}^{\infty} \frac{1}{r(r+1)}$.

a. 1

b. 2

c. 1/2

d. ∞

Section 4: Maclaurin Series & Higher-Order Approximations

10 State the general Maclaurin series formula for a function $f(x)$ expanded around $x = 0$.

a. $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$

b. $f(x) = f(0) + f'(0)x + f''(0)x^2 + f'''(0)x^3 + \ldots$

c. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(x)\frac{x^n}{n!}$

d. $f(x) = f(1) + f'(1)(x-1) + \ldots$

11 Find the first three non-zero terms in the Maclaurin series of $\ln(1+x)$.

a. $x - \frac{x^2}{2} + \frac{x^3}{3}$

b. $x + \frac{x^2}{2} + \frac{x^3}{3}$

c. $x - x^2 + x^3$

d. $1 + x - \frac{x^2}{2}$

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12 Find the coefficient of x^3 in the Maclaurin series of $e^x \sin x$.

a. $\frac{1}{3}$

b. $\frac{1}{6}$

c. $-\frac{1}{3}$

d. 0

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